User Manual
ASSA ABLOY High speed door

ASSA ABLOY

ASSA ABLOY Entrance Systems

The global leader in
door opening solutions



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About this Manual

Purpose of user manual HSD



All users and owners of the high speed door must read, understand and obey the information and instructions in this manual. Failure to do so may result in damage to, or failure of the equipment, and possible injury to persons.

This manual contains information and user instructions for a high speed door. When information or instructions are applicable to all the methods of operation or models, there are no operation types or model numbers in the title. When information or instructions are applicable to specific methods of operation or models, the applicable operation type or model numbers appear in the title.
To identify the model installed with your high speed door, refer to the descriptions in Chapter 1.

About ASSA ABLOY Entrance Systems

Solutions that support your business



ASSA ABLOY Entrance Systems is the world's most complete supplier of entrance automation solutions. With a holistic approach to the flow of people, goods and vehicles, we create solutions with the best possible balance of cost, quality and lifetime performance. Today we have gathered a strong portfolio of global brands under a single roof, preserving the wealth of expertise of each. Besam, Crawford, Megadoor and Albany are brands that are market leaders in their fields since decades, and have a strong history of proven solutions and extraordinary service know-how. Our solutions form a total offering for the front, back and interior of your building. Our combined approach means we understand the full range of challenges you face, and allows us to provide the best solution for your total entrance needs. Learn more about ASSA ABLOY Entrance Systems at www.assaabloyentrance.com.

Our relationship continues

As your entrances are part of your business flow, there's every reason to keep them working well. ASSA ABLOY Entrance Systems offers you a maintenance and modernization expertise to rely on.
Our Maintenance Programs and Modernization Services for entrance automation is backed by a extensive expertise for all types of pedestrian- and industrial door and docking systems, independent of brand. At your disposal is a team of dedicated expert technicians, proven through decades of maintenance, service and satisfied customers.

Maintenance of your high speed door

To make sure your ASSA ABLOY door continues to operate with maximum reliability, safety and performance, it is necessary to do regular maintenance as specified in this user manual and in the logbook supplied with your ASSA ABLOY door. This maintenance must be done by qualified service engineers trained to recognise possible problems, and to replace parts that have a defined life expectancy.

Your Local Service Centre

It is recommended that you contact your local ASSA ABLOY Entrance Systems Service Centre for details about ASSA ABLOY Entrance Systems Service Agreements available for your high speed door.

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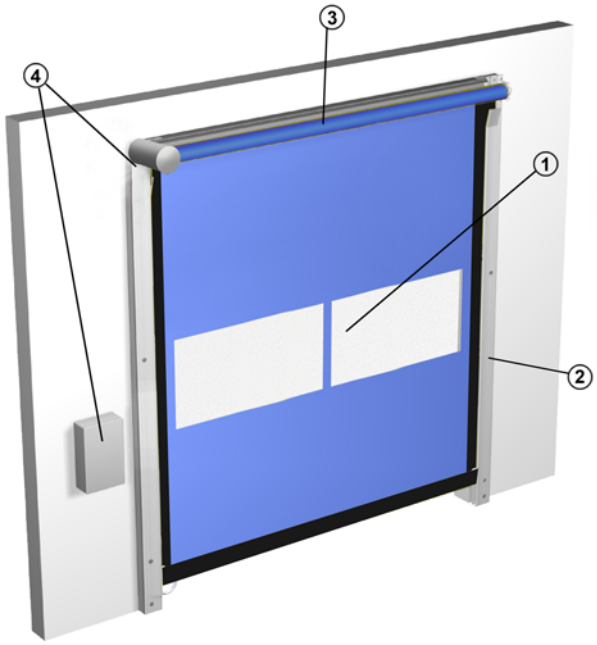
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1. Introduction

This document applies to the following models: HS8010P/PL, HS8020P/PL, HS9010P/PL/PEM/PFR, HS8110P, HS9110P, HS9020G/GAT/GHY, HS9030G/GAT/GEM/GHY.
The high speed door is a curtain door designed for either interior or exterior operations. It protects the environment against draughts, humidity, dust and dirt. With fast opening and closing speed, the door improves the traffic flow, provides employee comfort, and saves energy.



The ASSA ABLOY high speed door has 4 primary parts:

1) Door curtain
2) Side columns
3) Header box
4) Operating system

1.1 Door curtain

1.1.1 Construction

The door curtain is constructed from one single piece of PVC fabric. The door curtain rolls up above the door opening and requires little space.

Top

The top of the fabric is connected to a fabric roll, located in the header box above the door opening.

Bottom

The flexible bottom edge of the door curtain does not contain any stiffeners, making the door curtain completely safe as the curtain moulds itself in case a person is trapped below the closing door.

Side

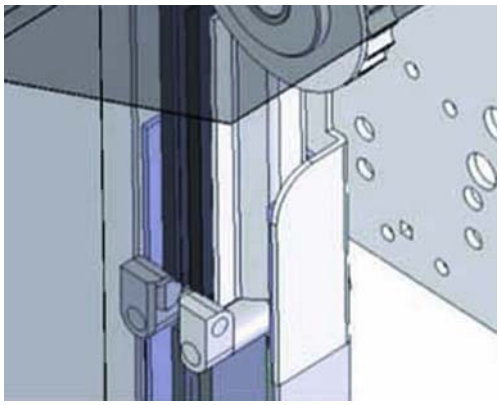
The left and right sides of the door curtain are constructed with unique bearing strings that keep the door curtain tensioned under wind load, but minimises the friction of the door curtain through the side columns. If the door is hit by a vehicle, a number of these bearing strings are pulled out of the side column. The self repair function acts as a zipper to put the bearing strings back in the side column.

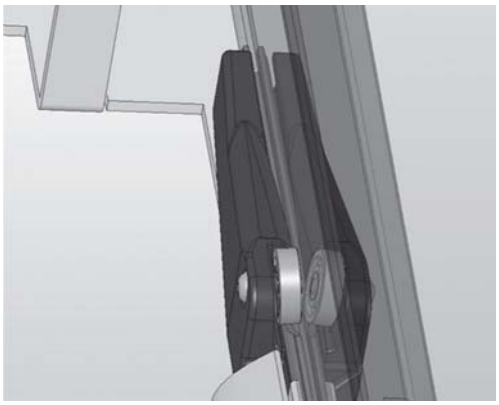
Emergency Exit curtain

The door curtain for an emergency exit is constructed of 3 sections of PVC fabric. These 3 sections are held together with 2 zippers which are protected by a protective cover.

1.1.2 Self repair system

The high speed doors are equipped with an automatic repair system. If a door is hit by a vehicle during operation, the resistant door curtain absorbs the impact and releases itself from its side guides without damage. The door reinserts itself automatically within the next open and close cycle. This unique feature makes the door crash-resistant without sustaining damage, reducing production downtime and maintenance.





1.2 Side columns

The side columns guide the door curtain up and down. This guidance is a plastic-to-plastic connection, which makes lubrication essential.

1.3 Fabric roll

The fabric roll is installed in the header box above the door curtain. Its function is to roll up or down the door curtain by means of the operating system. The door curtain is tensioned by its own fabric weight.



1.4 Operating system

1.4.1 General

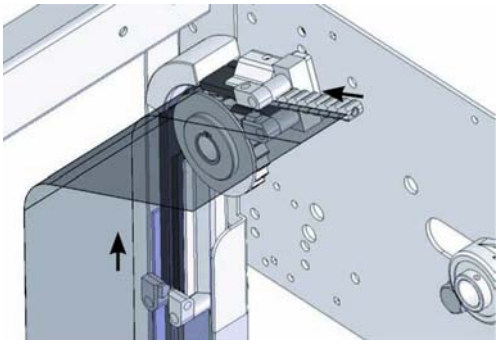
The ASSA ABLOY high speed door is always operated electrically. The operating system is a combination of an operator and a control unit. The operator opens the door with an electric engine. The door closes under the weight of the door curtain. The operator secures a safe closing speed.

1.4.1.1 Gravity system

The gravity high speed doors have some ballast in the bottom seal. The door closes under the weight of the door curtain. This system is suitable for inside doors, under a limited pressure difference and limited dimensions.

1.4.1.2 Door drive system

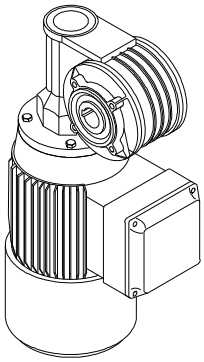
The high speed doors are equipped with a unique door drive system. This system eliminates the need for ballast in the door curtain or tension straps. This gear driven system consists of a pinion on the drive shaft that forces the lateral retaining straps up or down the tracks. This ensures that even under high pressure difference the door will be fully closed.



1.4.2 Operator

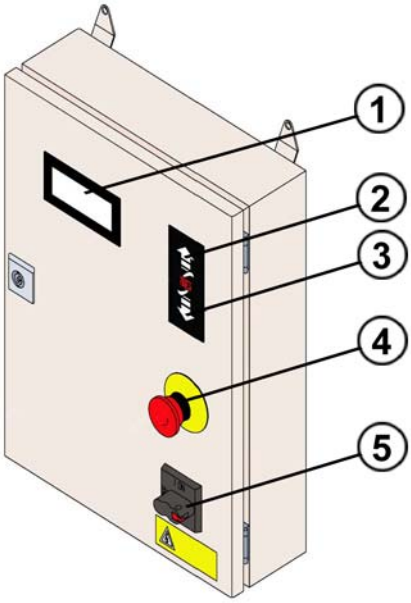
Exceptional reliability and smooth operation is ensured by a motor driven by a frequency inverter. This technology ensures a soft start and stop, which increases the longevity of the motor considerably. It also allows faster opening/closing speed. This motor delivers reliable operations around-the clock. The operator is always combined with a control unit.

The operator drives the fabric roll to open or close the door. In case of a main supply failure, the operator can be disconnected and the door can be opened or closed manually using the hand crank.



1.4.3 Control unit

The control unit is installed beside the door. It has impulse UP and DOWN buttons, an emergency stop mushroom button and a mechanical mains switch.



1) Display
2) UP button
3) DOWN button
4) Emergency stop
5) Mechanical main switch

1.5 Safety Provisions

1.5.1 Bottom edge

The bottom edge is a flexible bag that contains a wireless edge activator. When the door is hit by a vehicle, the curtain folds around the object and the contacts within the bottom edge make a closed circuit. The wireless signal is then sent to the operator to stop the door movement immediately.



2. Safety

2.1 General safety statements

The ASSA ABLOY high speed door has been designed to meet all operational and safety requirements in the European Directives and the standards issued by the European Standardization Committee, CEN.

2.2 Safety instructions

- Only authorized persons are permitted to operate the door.
- Read and fully understand the safety instructions before you operate or do maintenance on the door.
- Always obey local health and safety regulations when you operate or do maintenance on the door.

 Make sure there are no persons or equipment in the working area of the door before you operate the door.

- Do not remove or immobilize safety equipment installed on or near the door.
- Do not operate the door if it has any defects. Report defects to the supervisor immediately.
- Electrically disconnect and immobilize the dock leveller before you do maintenance on the dock leveller.
- Do not operate the door after the date of the next scheduled maintenance. The date of the next scheduled maintenance is shown in the logbook.
- Do not modify the door. Unauthorized modification can cause danger to people and affect the function and safety of the door.
  Unauthorized modification of the door will void the warranty.
- Do not use the door if the safety edge is not functioning properly.
- Do not attempt to adjust or disassemble any part of the high speed door.

2.3 Safety Symbols Used in this Manual

The following safety symbols are used in this manual:

 **Indicates a general warning**

 **Indicates an electrical hazard**

2.4 Safety Procedures

2.4.1 Emergency Stop Procedure

1) Press the STOP button to stop the door.
2) When the area is safe:
   1) To reset the emergency stop button:
      Carefully push the emergency stop button and turn it clockwise until it is reset. The door can now be operated normally again.
   1) Press the ⬆ or ⬇ button to operate the door.

2.4.2 Open/Close Door During Electrical Supply Failure

Do not lift the brake when the door is open and the hand crank is not installed. The door will close very quickly and may cause injury or damage.

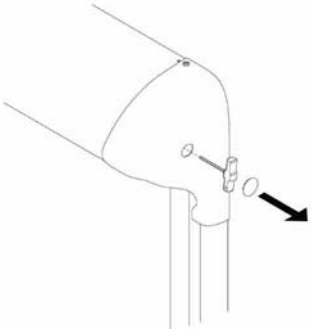
In the case of a UPS system, be prepared that the door will open quickly the moment the electrical current is interrupted.

When the hand crank is inserted into the mechanical unit, the control voltage is automatically disconnected.

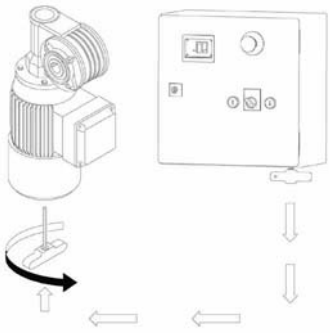
If the electrical power fails, a brake automatically locks the operator so that the door does not fall down (HS9030G series only).

1) Disconnect the main electrical supply from the door.
2) Lock the main switch (using padlock).
3) Wait at least 15 minutes after locking the main switch before starting work. This is required for discharging the capacitors of the frequency inverter.
4) Take the hand crank from its support under the control box
   - (HS9030G series) remove rubber cap and insert the hand crank it into the opening at the back the motor.
   - (all other series) insert the hand crank it into the opening under the motor.

HS9030G series



other series



5) Lift the brake lever (HS9030G) and simultaneously (all) turn the hand crank to open or close the door.
   HS8010P:Clockwise
   HS8010PL:Clockwise
   HS8020P:Clockwise
   HS8020PL:Clockwise
   HS8110P:Clockwise
   HS9010P:Clockwise
   HS9010PL:Clockwise
   HS9020G:Counter clockwise
   HS9030G:Counter clockwise
   HS9110P:Clockwise
6) Lower the brake lever (HS9030G).
7) Remove the hand crank and return it to its position under the control box.

The door can be operated normally when the electrical supply is available again.

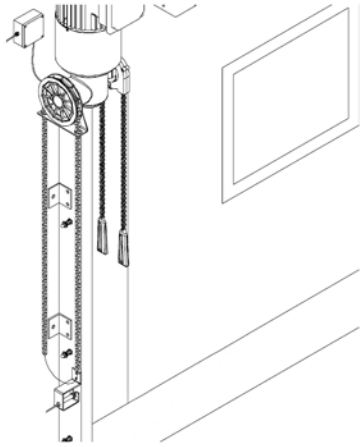
2.4.2.1 Manual Chain Hoist (Optional for HS9010P, HS9110P, HS8010P, HS8020P and HS8110P series only)

Pull red handle down to interrupt safety circuit (same function as the emergency stop).

This prevents the motor from starting unexpectedly during manual operation.

1) Pull the chain in either direction, opening or closing the door.
2) Disconnect the chain hoist with the green handle and reset the door by pushing the "open" button in the control panel.

Operating manual hoist



3. Operating Instructions

3.1 Daily procedures

3.1.1 Daily Start Procedure



Do not stand under the door when you test the safety edge.

1) Examine the door for damage.
2) If damage is found:
   1) Put suitable warning signs around the door.
   2) Contact the applicable person.
3) Open and close the door four or five times and make sure:
   - The door operates correctly (see opening and closing instructions)
   - The door operates smoothly without any unusual noise.
4) Test the edge activator as follows:
   1) Press and release the button and make sure the door moves to the fully open position.
   3) Press the button.
   4) When the door starts to move down, tap the bottom of the edge activator with your hand.
   5) Make sure the door reverses immediately and moves to the fully open position.
   6) Press the button to close the door.

3.1.2 Open and close the door

3.1.2.1 Open the door



Make sure there are no persons or equipment in the working area of the door before you operate the door.

1) Make sure the Daily Start Procedure has been done.
2) Press and release the button and make sure the door moves to the fully open position.

3.1.2.2 Close the door



Make sure there are no persons or equipment in the working area of the door before you operate the door.

1) Press and release the button and make sure the door fully closes.

3.1.2.3 Stop the door



Make sure there are no persons or equipment in the working area of the door before you operate the door.

1) Press the red emergency stop button to stop the door. After the emergency stop button has been pushed, it must be reset for the door to be operational again.
2) To reset the emergency stop button:
   Carefully push the emergency stop button and turn it clockwise until it is reset. The door can now be operated normally again.

3.1.2.4 Auto/Manual switch (Optional)

1) Press push button and hold until closed. The photocell and the wireless detector remain active.
2) Press "open" button once to open the door again.

3.1.3 Automatic self-repair function

If the door is hit while moving and the sides break away from the side columns, the door leaf will travel to the top while relocating in the side columns. After the door leaf again reaches the bottom, check to make sure the door leaf is back in place.

When the door is hit while the fabric is on the ground and the sides break away

1) Lift the door leaf to relocate the fabric in the side columns.
2) Lower the door leaf to the bottom and check to make sure the it is correctly back in place.

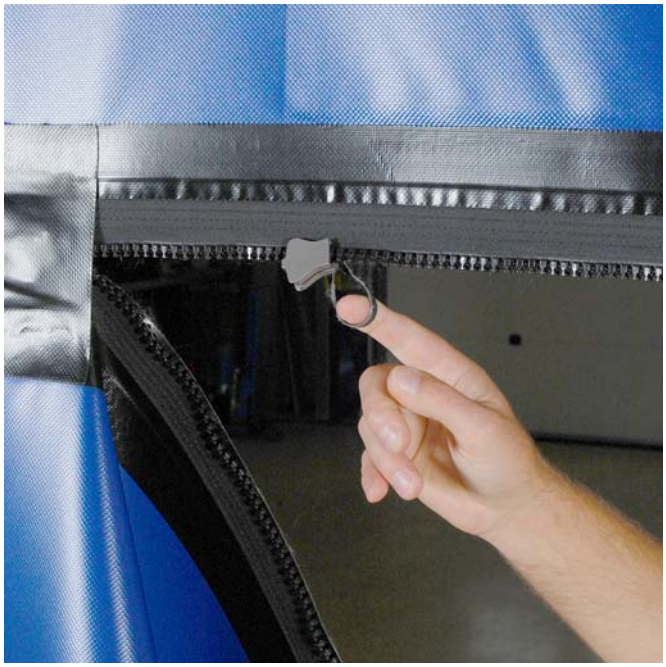
Contact your local ASSA ABLOY Entrance Systems Service Center if damage is found.

3.2 Procedure for repairing door panel

In case of emergency open the door.



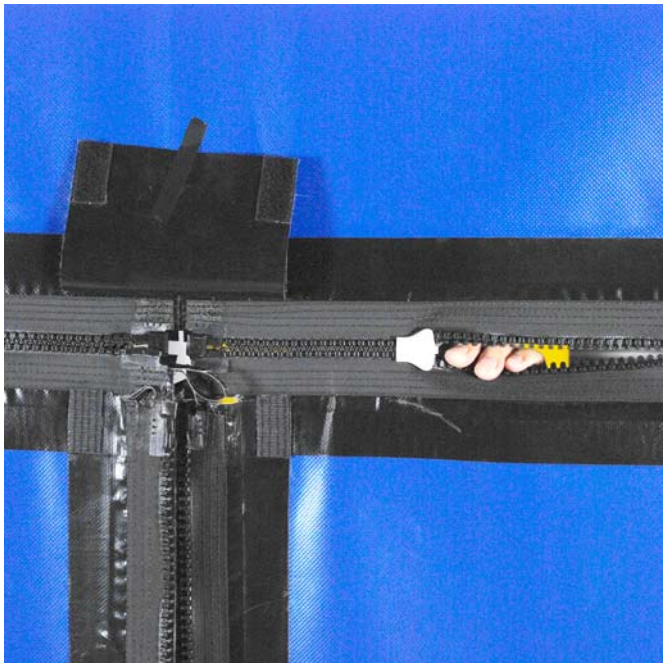
Pull the horizontal zipper connectors to the middle of the door.



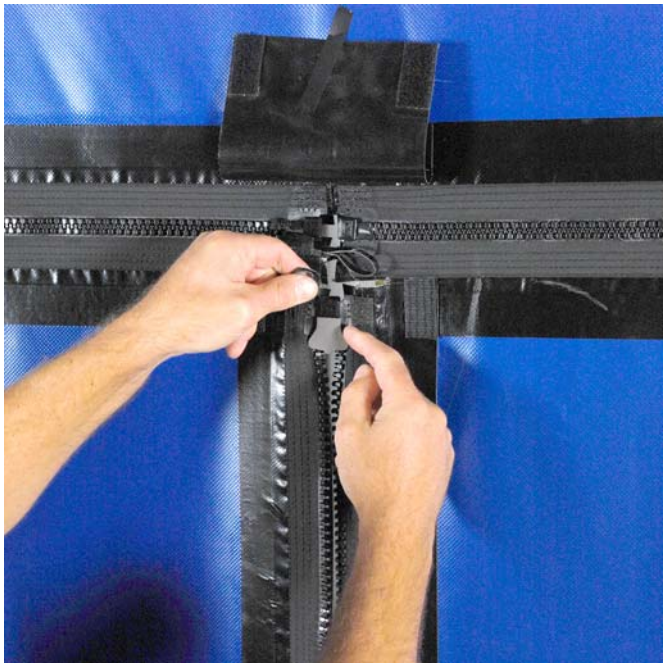


Close the horizontal zippers to the end.





Connect the vertical zippers and close it until the red marking.





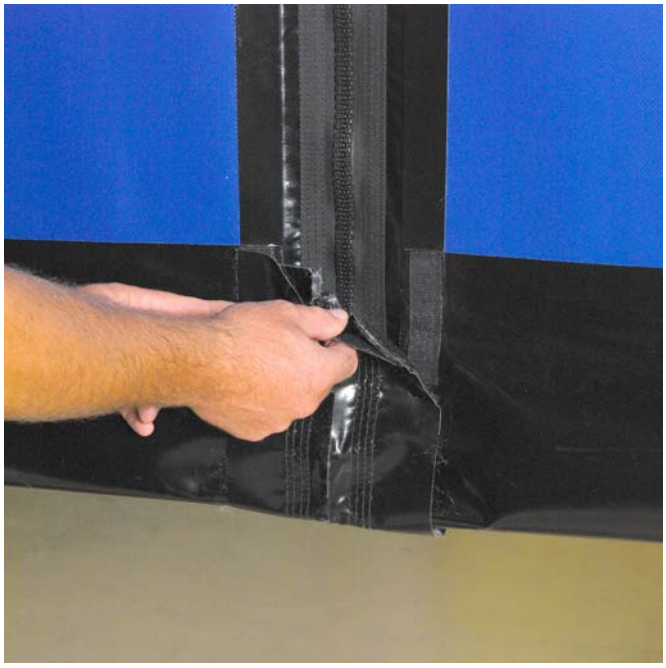
Send the door up halfway with the up-button.



Close the vertical zipper completely.



Close the vertical zipper completely.



Door is ready for use.



4. Maintenance

In this chapter you find the maintenance tasks that you as a user must perform. With the exception of inspections and maintenance tasks explicitly described in this manual as being performable by the user, all other maintenance tasks, as described in the Logbook are rated advanced and can only be carried out safely by a competent person. Contact your local Service Centre.

4.1 Preventive maintenance schedule

Frequency	Part	Tasks
Daily	Door complete	Do the Daily Start Procedure.
Weekly	HS9010PFR only	Remove ice formation around top and bottom seals
Every three months	Door leaf	Clean the door leaf.
Every six months or 50.000 door cycles	Door complete	Service visit

4.2 Preventive maintenance procedures

4.2.1 Clean the door leaf



Do not use the door leaf or the side columns to support a ladder when you do maintenance on a door. Always use ladders as specified in local health and safety instructions.



Do not use a high pressure water hose to clean the door.

1) Do the Daily Stop Procedure.
2) Use a soft clean brush and mild detergent to clean the inside and outside of the door fabric.
3) If damage is found, contact the local service centre for repairs.

4.2.2 Remove Ice Formation (HS9010PFR only)



Do not use the door leaf or the side columns to support a ladder when you do maintenance on a door. Always use ladders as specified in local health and safety instructions.



Do not use a high pressure water hose to clean the door.

1) Do the Daily Stop Procedure.
2) Remove ice formation in the top and bottom seals.
3) If damage is found, contact the local service centre for repairs.

5. Tests and Adjustments

5.1 Test the safety edge



Make sure there are no persons or equipment in the working area of the door before you operate the door.



Do not stand under the door when you test the safety edge.

1) Test the edge activator as follows:
    1) Press and release the ⬆ button and make sure the door moves to the fully open position.
    2) Press the ⬇ button.
    3) When the door starts to move down, tap the bottom of the edge activator with your hand.
    4) Make sure the door reverses immediately and moves to the fully open position.
    5) Press the ⬇ button to close the door.
2) If the safety edge is not functioning properly:
    1) Do not use the door.
    2) Put suitable warning signs around the door.
    3) Contact ASSA ABLOY Entrance Systems Service.

6. Troubleshooting

6.1 Introduction

This chapter contains troubleshooting information for users of an high speed door. If a fault is not described in this chapter contact your local ASSA ABLOY Entrance Systems Service Centre for assistance.

6.2 Door does not go up or down correctly

Possible Cause	Solution
Damage to door or side columns	Contact ASSA ABLOY Entrance Systems service
Malfunction of safety edge	Make sure the door leaf is engaged correctly in the columns.
No main electrical supply	Switch on main electrical supply
Emergency stop locked	Unlock emergency stop
Obstacle in the door opening	Remove object(s)

6.3 Error code displayed on control unit

Because an error code may appear when there is not a fault, do the following procedure when an error code appears:

1) Switch off the supply voltage.
2) Wait for 5 seconds.
3) Switch on the supply voltage.
4) If an error code is still displayed refer to the error code list.

6.3.1 Error code list

The following tables give the error codes that can be displayed on the control unit. It also describes the cause of the error and the action necessary to remove the error.

Important messages. Call ASSA ABLOY Entrance Systems service !!

Code	Description	Solution
I.080	Maintenance required	Call ASSA ABLOY Entrance Systems service
F.360	Safety edge is short circuited	Call ASSA ABLOY Entrance Systems service
F.363	Safety edge interrupted	Call ASSA ABLOY Entrance Systems service

Symptom: Door does not go up or down

Code	Description	Solution
F.201	Emergency button is stuck in lid	Turn button a quarter and pull
F.211 and F.212	External emergency stop was activated or hand crank is not in place	Release emergency stop and place hand crank in control unit

General messages (no needed action/only information)

Code	Description	Remark
STOP	Stop activated	Wait for next incoming command
Eu	Door in end postition CLOSED	
Eo	Door in end position OPEN	
ZUF	Closing active	
AUF	Opening active	
Hd'	Door in manual (deadman) mode	
Au'	Automatic mode	Indicates change from manual to automatic
Hc'	Semi-automatic mode	Indicates change from manual to semi-automatic
I.310	Door 2 not closed (if installed)	Only with "airlock function"

All other codes are messages of normal functioning (no needed action/only information).

7. Extract of Declaration of Conformity / Performance

We, ASSA ABLOY Entrance Systems AB
Lodjursgatan 10
SE-261 44 Landskrona
Sweden,

declare under our sole responsibility that the type of equipment:
power operated doortype ASSA ABLOY HS9010P, HS9010PEM, HS9010PFR, HS9010PL, HS9020G, HS9020GAT, HS9020GHY, HS9030G, HS9030GAT, HS9030GHY, HS9030GEM, HS9110P, HS8010P, HS8010PL, HS8020P, HS8020PL and HS8110P with performance levels and electrical drive unit as identified in the logbook provided with it, is in compliance with the following directives:

2006/42/EC Machinery Directive (MD)
2004/108/EC Electromagnetic Compatibility Directive (EMCD)

Harmonized European standards which have been applied:
EN 13241-1 EN 61000-6-2 EN 61000-6-3 EN 60204-1 EN 60335-1
Other standards or technical specifications, which have been applied:
EN 60335-2-103
EC type examination or certificate issued by a notified or competent body (for full address, please contact ASSA ABLOY Entrance Systems AB) concerning the equipment:

BBRI: 1136

The manufacturing process ensures the compliance of the equipment with the technical file.
Compilation of technical file:
Charles Fauquette
ASSA ABLOY Entrance Systems AB
Lodjursgatan 10
SE-261 44 Landskrona
Sweden

ASSA ABLOY Entrance Systems is a leading supplier of entrance automation solutions for the efficient flow of goods and people. Building on the long-term success of the Besam, Crawford, Albany and Megadoor brands, we offer our solutions under the ASSA ABLOY brand. Our products and services are dedicated to satisfying end-user needs for safe, secure, convenient and sustainable operations.
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